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Lecture 11A
Before-Tutorial-Session
Tut10

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Consider $\mathbb{R}^3$. Which of the following is an orthonormal basis?

1. $\left\{(1,0,1), \left(-\frac{1}{2}, 1, \frac{1}{2}\right), \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}\right)\right\}$

2. $\left\{\frac{1}{\sqrt{2}}(1,0,1), \frac{2}{\sqrt{6}}(-1,2,1), \frac{1}{\sqrt{3}}(1,1,-1)\right\}$

Let’s do these questions today. Tut10, Q3+Q6

Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(a) Apply the Gram-Schmidt process to $\mathcal{B}$ to obtain an orthogonal basis for $\mathbb{R}^3$.

(b) Normalize the vectors in the basis in part(a) to obtain an orthonormal basis $\mathcal{A}$ for $\mathbb{R}^3$.

(c) Let $v = (1,1,2)$. What is $[v]_{\mathcal{A}}$?

(d) Let $\boldsymbol{M} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 3 \end{bmatrix}$, and $\boldsymbol{Q}$ be the matrix whose columns are basis vectors in the orthonormal basis $\mathcal{A}$. Find a matrix $\boldsymbol{R}$ such that $\boldsymbol{M} = \boldsymbol{QR}$.

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(a) $\langle p(x), q(x)\rangle = p(0)q(0)$.

(b) $\langle p(x), q(x)\rangle = p(0)q(0) + p(1)q(1)$.

(c) $\langle p(x), q(x)\rangle = \int_{-1}^{1} p(x)^2 q(x)\, dx$.

Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(a) Apply the Gram-Schmidt process to $\mathcal{B}$ to obtain an orthogonal basis for $\mathbb{R}^3$.

What is $\boldsymbol{v}_1$?

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Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(a) Apply the Gram-Schmidt process to $\mathcal{B}$ to obtain an orthogonal basis for $\mathbb{R}^3$.

What is $\boldsymbol{v}_2$? $\left[\ \text{HIDDEN}\ \right]$

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Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(a) Apply the Gram-Schmidt process to $\mathcal{B}$ to obtain an orthogonal basis for $\mathbb{R}^3$.

What is $\boldsymbol{v}_3$?

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Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(b) Normalize the vectors in the basis in part(a) to obtain an orthonormal basis $\mathcal{A}$ for $\mathbb{R}^3$.

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What is $\mathcal{A}$?

Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(d) Let $M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 3 \end{bmatrix}$, and Q be the matrix whose columns are basis vectors in the orthonormal basis $\mathcal{A}$. Find a matrix R such that $M = QR$.

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Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1,0,1),(0,1,1),(1,3,3)\}$.

(c) Let $v = (1,1,2)$. What is $[v]_{\mathcal{A}}$?

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(d) Let $\boldsymbol{M} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 3 \end{bmatrix}$, and $\boldsymbol{Q}$ be the matrix whose columns are basis vectors in the orthonormal basis $\mathcal{A}$. Find a matrix $\boldsymbol{R}$ such that $\boldsymbol{M} = \boldsymbol{Q}\boldsymbol{R}$.

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Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(a) $\langle p(x), q(x) \rangle = p(0)q(0)$.

(b) $\langle p(x), q(x) \rangle = p(0)q(0) + p(1)q(1)$.

(c) $\langle p(x), q(x) \rangle = \int_{-1}^{1} p(x)^2 q(x)\, dx$.

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

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(a) $\langle p(x), q(x)\rangle = p(0)q(0)$.

<u>Albert's solution</u>.

Consider the function $p(x) = x$.

Then $\langle p(x), p(x)\rangle = p(0)p(0) = 0$.

Since $p(x)$ is not the zero function and yet $\langle p(x), p(x)\rangle = 0$, the product is not definite.

Therefore, the product is NOT an inner product.

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

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(b) $\langle p(x), q(x)\rangle = p(0)q(0) + p(1)q(1)$.

Let's prove definiteness.

Our Task: $\langle p(x), p(x)\rangle$ **BLANK1** implies that $p(x)$ **BLANK2**

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

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(b) $\langle p(x), q(x) \rangle = p(0)q(0) + p(1)q(1)$.

Let's prove definiteness.

$$\langle p(x), p(x) \rangle = p(0)^2 + p(1)^2$$

Why?

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

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(b) $\langle p(x), q(x)\rangle = p(0)q(0) + p(1)q(1)$.

Let's prove definiteness.

$p(0)^2 + p(1)^2 = 0$ implies that $p(0) = p(1) = 0$

Why?

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(b) $\langle p(x), q(x)\rangle = p(0)q(0) + p(1)q(1)$.

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Let's prove definiteness.

Let $p(x) = a_0 + a_1 x$.

Why?

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(b) $\langle p(x), q(x) \rangle = p(0)q(0) + p(1)q(1).$

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Let's prove definiteness.

Let $p(x) = a_0 + a_1 x.$

So, $p(0) =$ BLANK1 and $p(1) =$ BLANK2

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(b) $\langle p(x), q(x)\rangle = p(0)q(0) + p(1)q(1).$

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Let’s prove definiteness.

Therefore, the product is an inner product.
Is this correct?

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(c) $\langle p(x), q(x)\rangle = \int_{-1}^{1} p(x)^2 q(x)\, dx.$

<u>**Albert's solution**</u>.

Consider the function $p(x) = q(x) = 1$ and $\lambda = 1024$.

Then $\langle 1024p(x), q(x)\rangle = \int_{-1}^{1} 1048576\, dx = 2097152.$

But $1024\langle p(x), q(x)\rangle = 1024 \int_{-1}^{1} 1\, dx = 2048.$

Since $\langle 1024p(x), q(x)\rangle \neq 1024\langle p(x), q(x)\rangle$, the product is NOT homogeneous in first slot.

Therefore, the product is NOT an inner product.